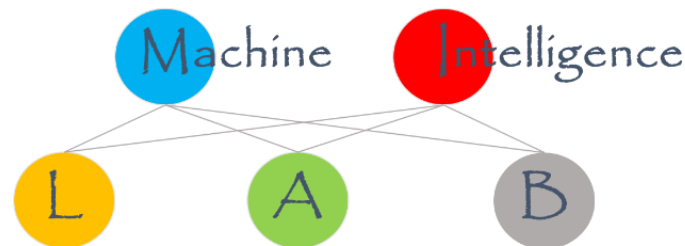




Object Detection – II

Mu Yadong

Wangxuan Institute of Computer Technology
Peking University

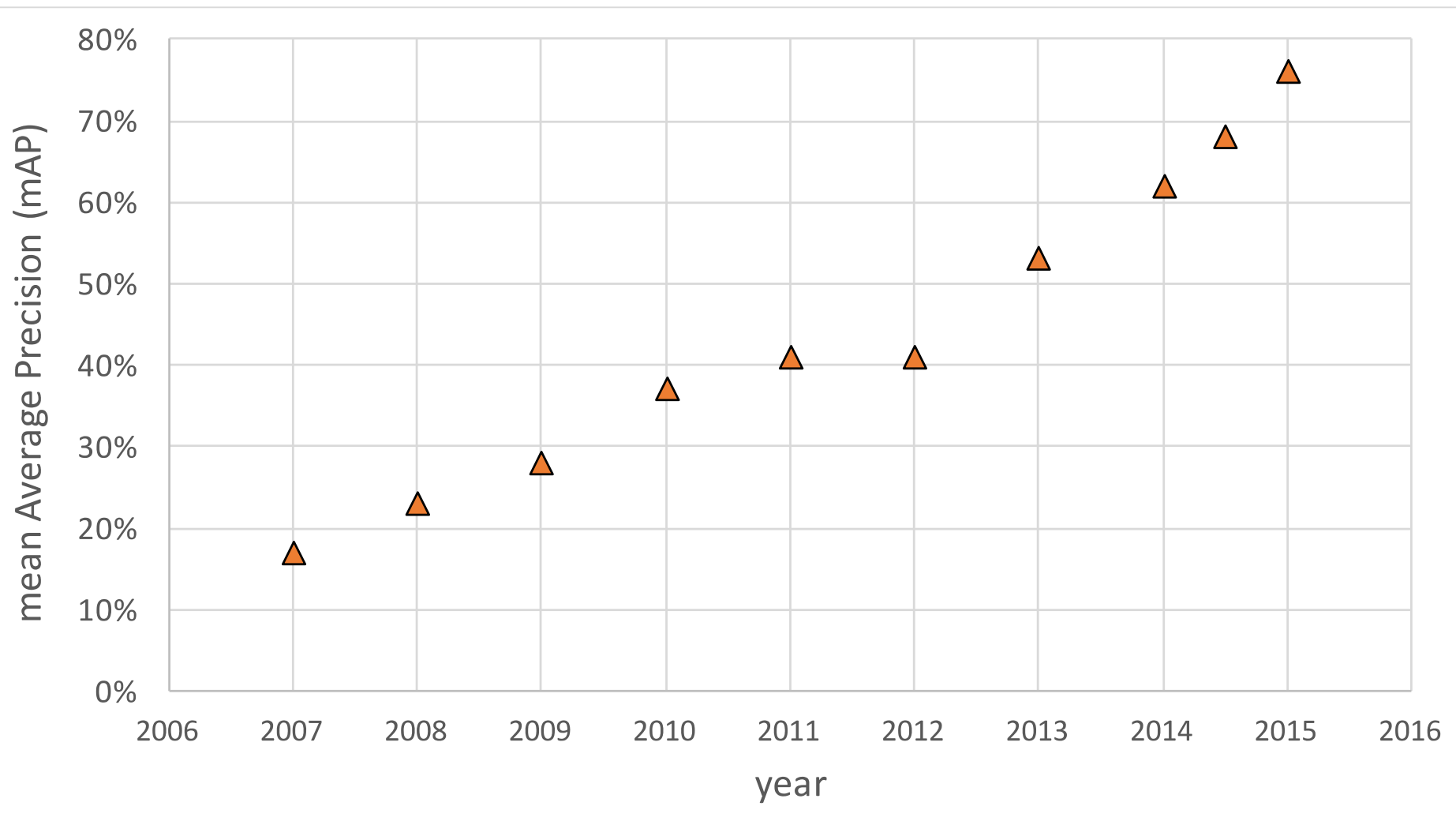


**Several slides are adapted from related courses or tutorials.
Internal use only. Please do not distribute the slides.**

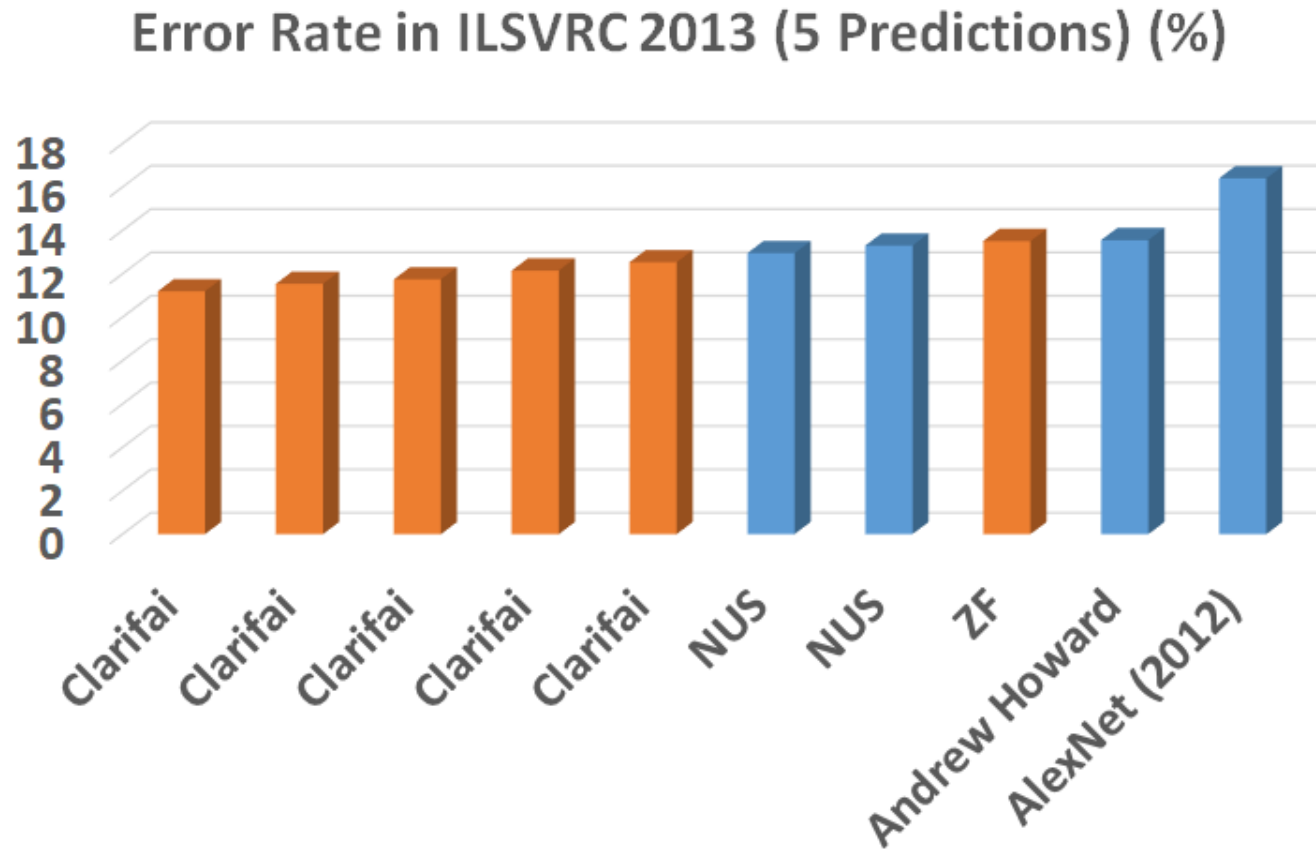
Outline

- **R-CNN, Fast R-CNN, Faster R-CNN**
- **YOLO, SSD**
- **Other Extensions**

Object detection progress



ZFNet

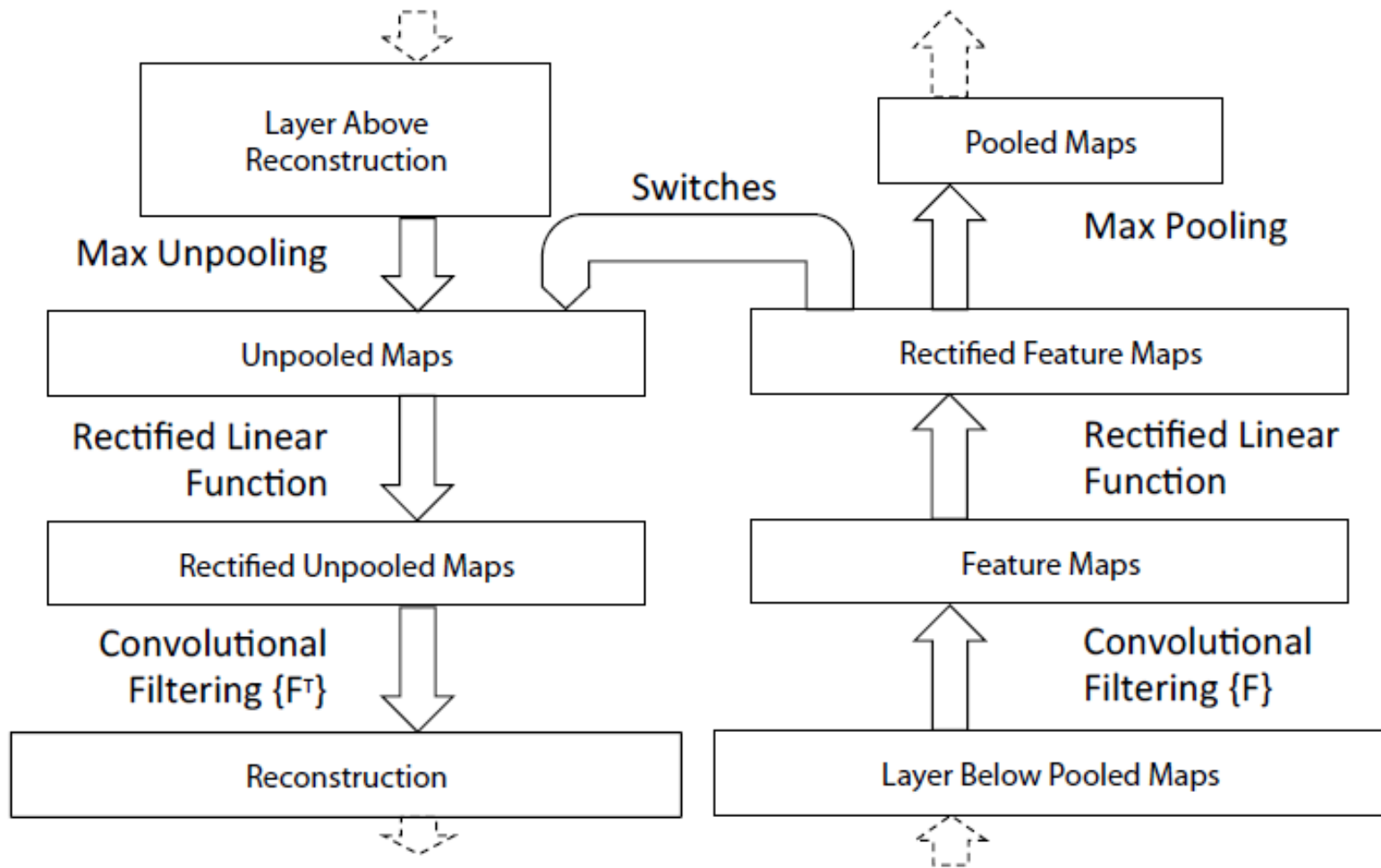


Matthew D Zeiler, Rob Fergus, Visualizing and Understanding Convolutional Networks, ECCV 2014

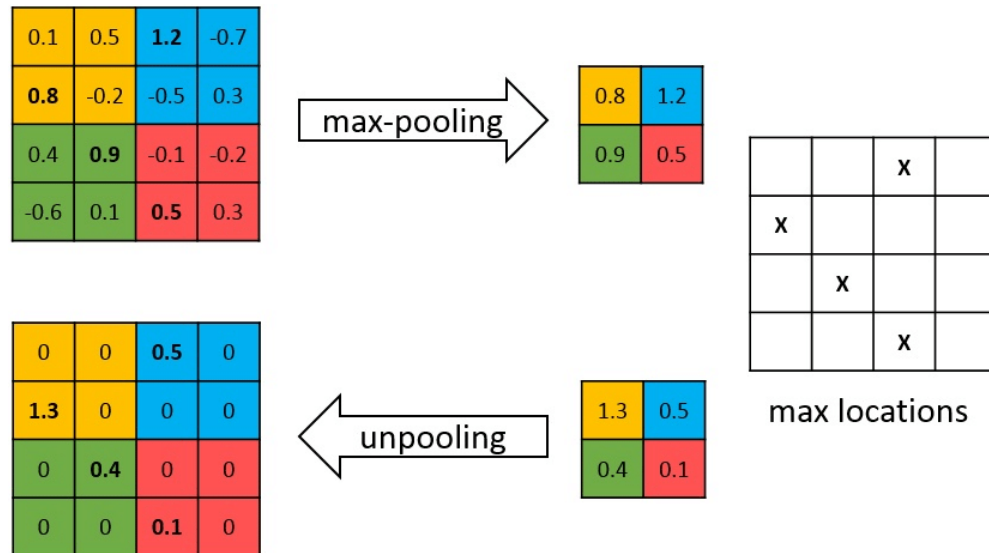
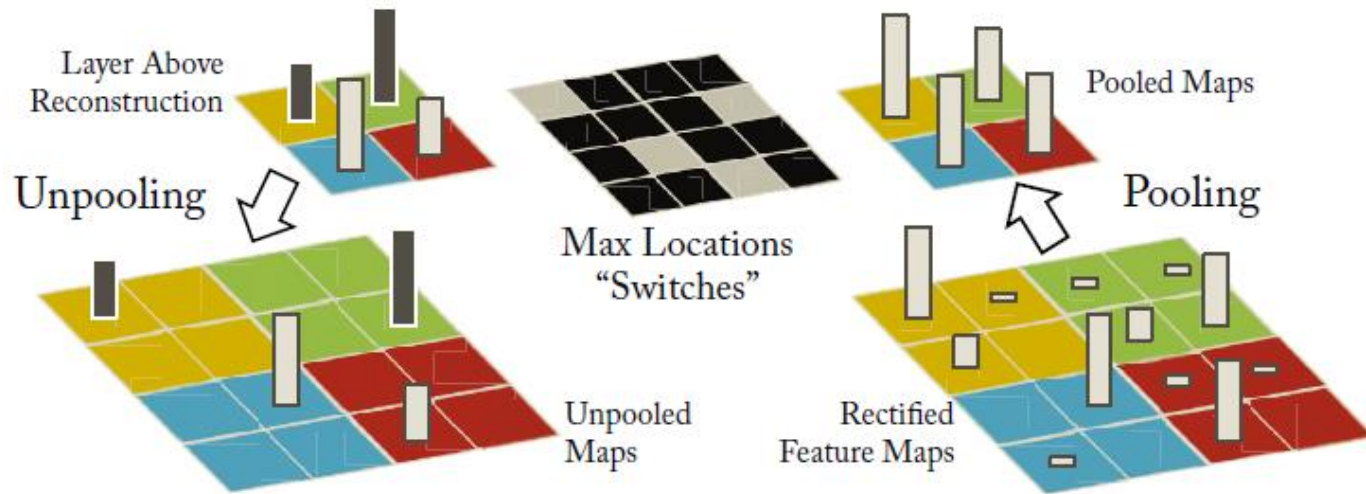
ZFNet

- a standard step in deep learning framework is to have a series of Conv > Rectification (Activation Function) > Pooling.
- To visualize a deep layer feature, we need a set of unpooling / deconvnet techniques to reverse the above actions such that we can visualize the feature in pixel domain (i.e., take an arbitrary neuron in some feature map, draw its corresponding pattern in the original image space).

ZFNet

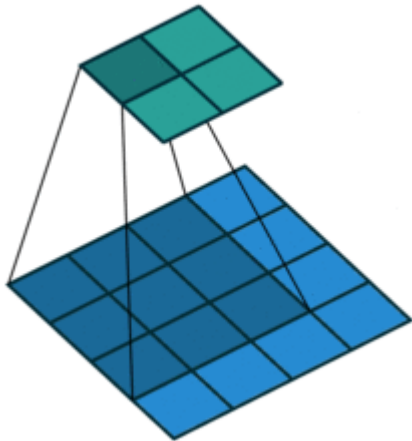


Un-Pooling

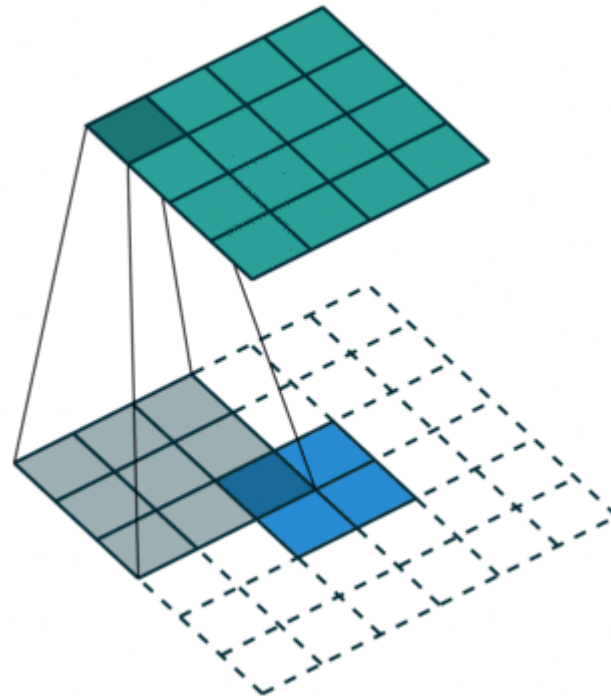


De-Conv

- Input : 4*4, kernel 3*3, output: 2*2
- Re-arrange input into 16-d (denote it as x), and output into 4-d (as y)
- $y = C * x$ (conv) and $x = C^T * y$ (deconv). Conv. kernel is horizontally / vertically flipped for deconv.



Conv (Blue is input, cyan is output)



Deconv (Blue is input, cyan is output)

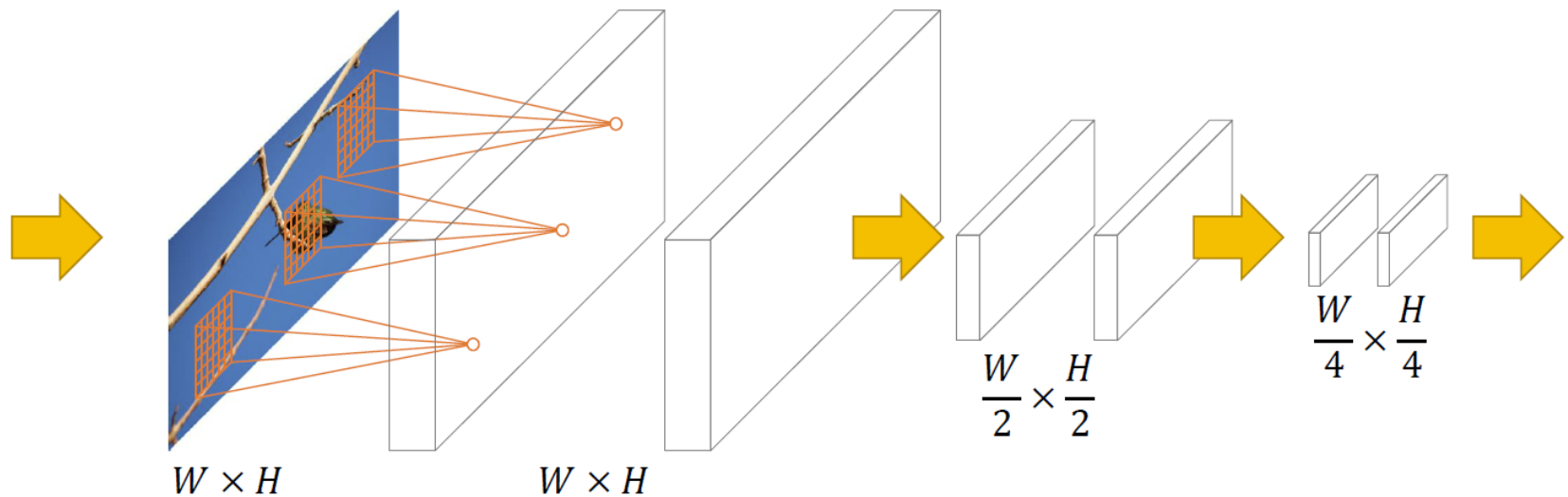
De-Conv

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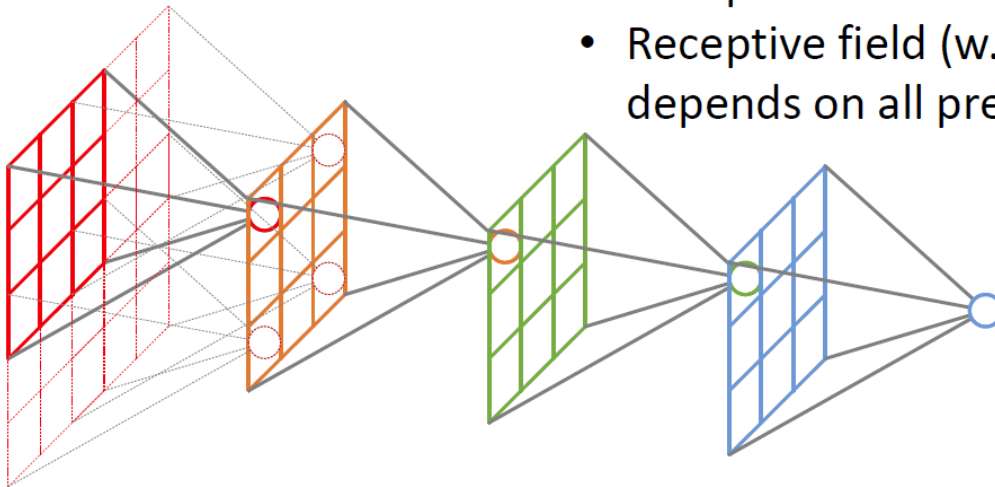
$$C = \begin{pmatrix} w_{0,0} & w_{0,1} & w_{0,2} & 0 & w_{1,0} & w_{1,1} & w_{1,2} & 0 & w_{2,0} & w_{2,1} & w_{2,2} & 0 & 0 & 0 & 0 & 0 \\ 0 & w_{0,0} & w_{0,1} & w_{0,2} & 0 & w_{1,0} & w_{1,1} & w_{1,2} & 0 & w_{2,0} & w_{2,1} & w_{2,2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & w_{0,0} & w_{0,1} & w_{0,2} & 0 & w_{1,0} & w_{1,1} & w_{1,2} & 0 & w_{2,0} & w_{2,1} & w_{2,2} & 0 \\ 0 & 0 & 0 & 0 & 0 & w_{0,0} & w_{0,1} & w_{0,2} & 0 & w_{1,0} & w_{1,1} & w_{1,2} & 0 & w_{2,0} & w_{2,1} & w_{2,2} \end{pmatrix}$$

Convolution

- Convolutional layers can be applied to images of any sizes, yielding proportionally-sized outputs



Receptive Field in Convolution

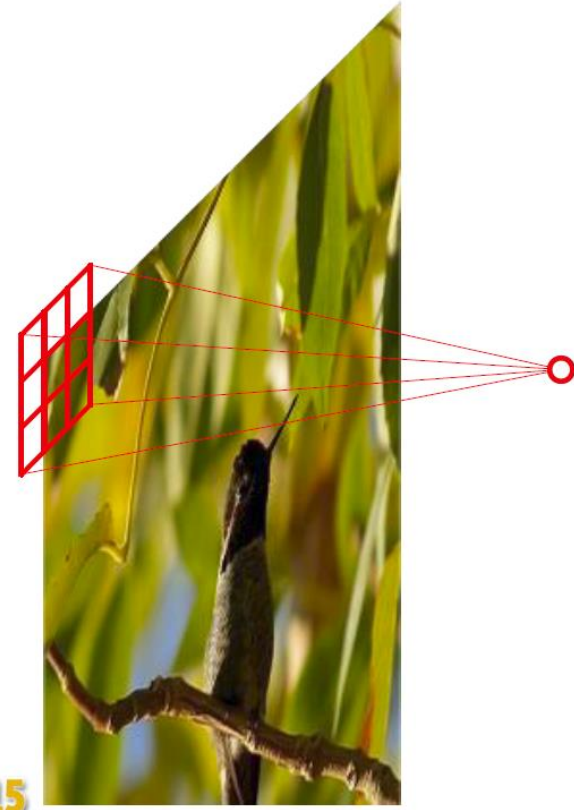


- Receptive field of the first layer is the filter size
- Receptive field (w.r.t. input image) of a deeper layer depends on all previous layers' filter size and strides

- **Correspondence** between a feature map pixel and an image pixel is not unique
- Map a feature map pixel to **the center of the receptive field** on the image in the SPP-net paper

Kaiming He, Xiangyu Zhang, Shaoqing Ren, & Jian Sun. "Spatial Pyramid Pooling in Deep Convolutional Networks for Visual Recognition". ECCV 2014.

Receptive Field in Convolution



How to compute the center of the receptive field

- A simple solution
 - For each layer, pad $\lfloor F/2 \rfloor$ pixels for a filter size F (e.g., pad 1 pixel for a filter size of 3)
 - On each feature map, the response at $(0, 0)$ has a receptive field centered at $(0, 0)$ on the image
 - On each feature map, the response at (x, y) has a receptive field centered at (Sx, Sy) on the image (stride S)

- A general solution

$$i_0 = g_L(i_L) = \alpha_L(i_L - 1) + \beta_L,$$

$$\alpha_L = \prod_{p=1}^L S_p,$$

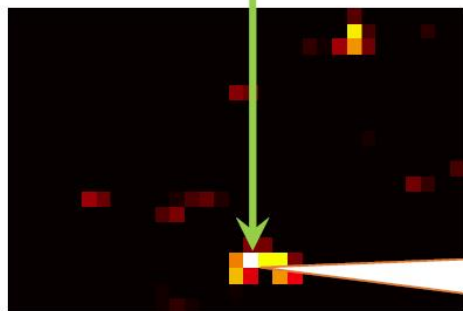
$$\beta_L = 1 + \sum_{p=1}^L \left(\prod_{q=1}^{p-1} S_q \right) \left(\frac{F_p - 1}{2} - P_p \right)$$

See [Karel Lenc & Andrea Vedaldi]
"R-CNN minus R". BMVC 2015.

Kaiming He, Xiangyu Zhang, Shaoqing Ren, & Jian Sun. "Spatial Pyramid Pooling in Deep Convolutional Networks for Visual Recognition". ECCV 2014.

Feature Maps = features and their locations

ImageNet images with **strongest** responses of this channel



one feature map of conv₅
(#55 in 256 channels of a model
trained on ImageNet)



Intuition of *this* response:
There is a “**circle-shaped**” object (likely a tire) **at this position**.

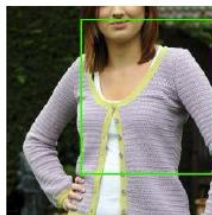
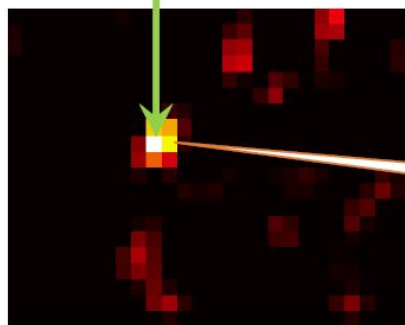
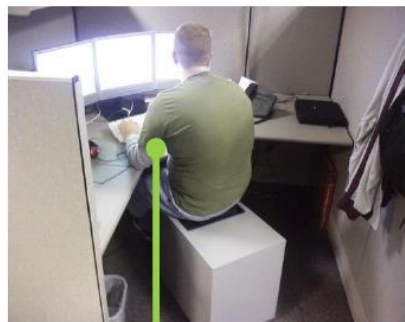
What

Where

Slides credit: Kaiming He, ICCV 2015 tutorial

Feature Maps = features and their locations

ImageNet images with **strongest** responses of this channel



Intuition of *this* response:

There is a “ **λ -shaped**” object (likely an underarm) **at this position**.

What

Where

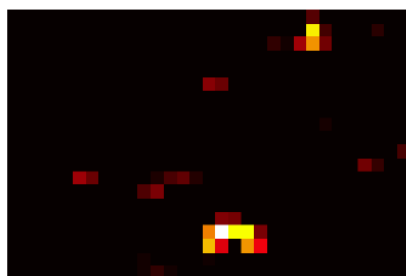
one feature map of conv₅
(#66 in 256 channels of a model
trained on ImageNet)

Feature Maps = features and their locations

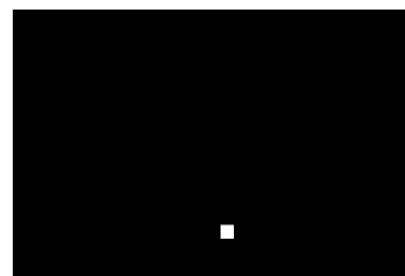
- Visualizing **one response** (by Zeiler and Fergus)



image



a feature map



keep one response
(e.g., the strongest)



Feature Visualization with ZFNet

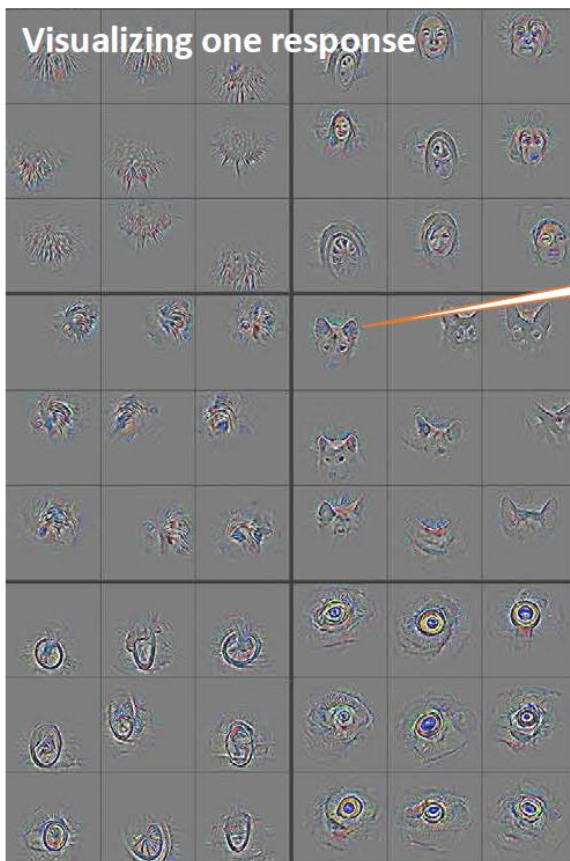


conv3



image credit: Zeiler & Fergus

Feature Visualization with ZFNet



conv5

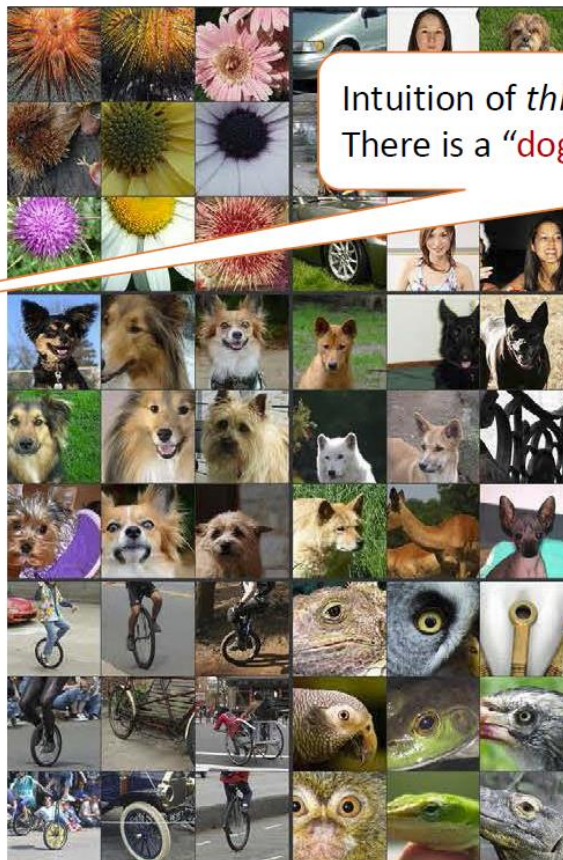


image credit: Zeiler & Fergus

Intuition of *this visualization*:

There is a “dog-head” shape *at this position*.

- **Location** of a feature: explicitly represents *where* it is.
- **Responses** of a feature: encode *what* it is, and implicitly encode finer position information –

finer position information is encoded in the channel dimensions (e.g., bbox regression from responses at one pixel as in RPN)

Matthew D. Zeiler & Rob Fergus. “Visualizing and Understanding Convolutional Networks”. ECCV 2014.

Region-based CNN Features

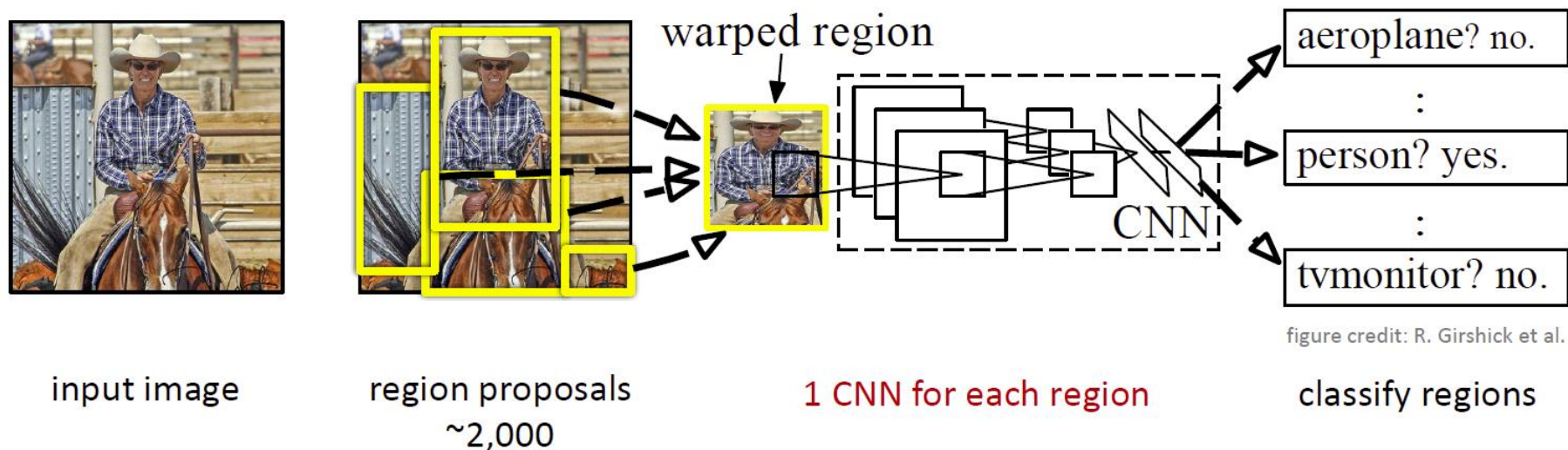


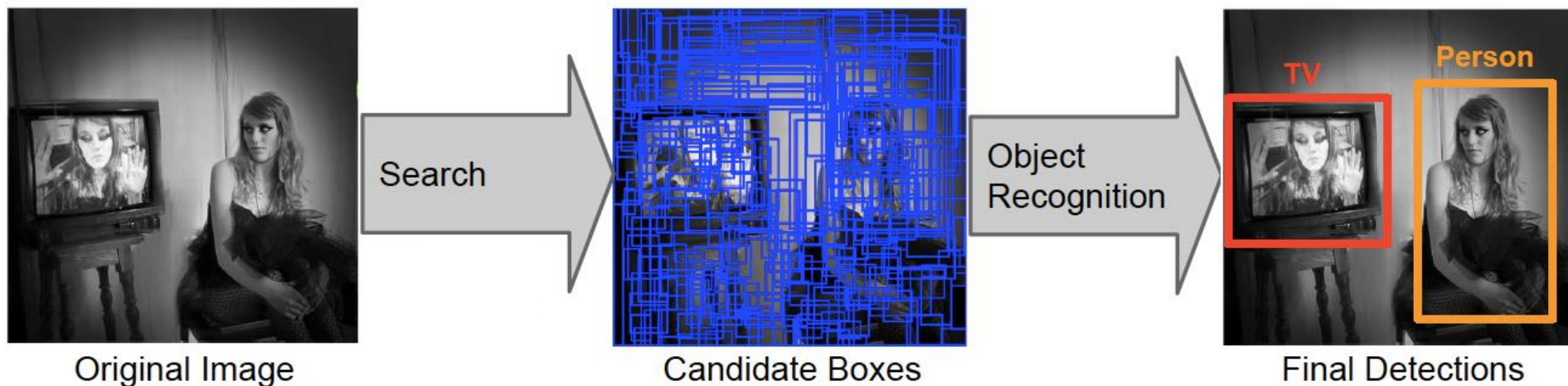
figure credit: R. Girshick et al.

R-CNN pipeline

R. Girshick, J. Donahue, T. Darrell, & J. Malik. "Rich feature hierarchies for accurate object detection and semantic segmentation". CVPR 2014.

Beyond sliding windows: Region proposals

- **Advantages:**
 - Cuts down on number of regions detector must evaluate
 - Allows detector to use more powerful features and classifiers
 - Uses low-level *perceptual organization* cues
 - Proposal mechanism can be category-independent
 - Proposal mechanism can be trained



Selective search: Basic idea

- Use hierarchical segmentation: start with small *superpixels* and merge based on diverse cues



J. Uijlings, K. van de Sande, T. Gevers, and A. Smeulders, Selective Search for Object Recognition, IJCV 2013

Region-based CNN Features

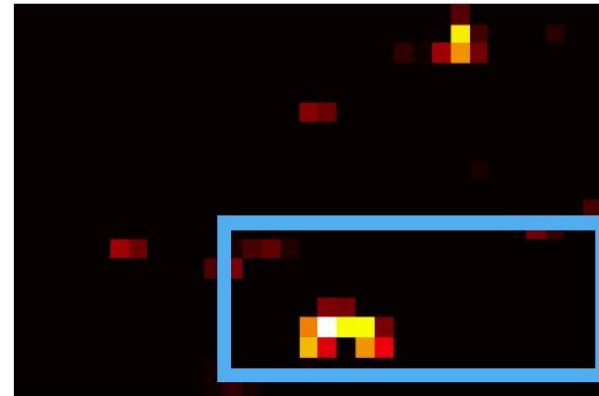
- Given proposal regions, what we need is a feature for each region
- R-CNN: cropping an image region+ CNN on region, requires 2000 CNN computations
- What about cropping feature map regions?



Regions on Feature Maps



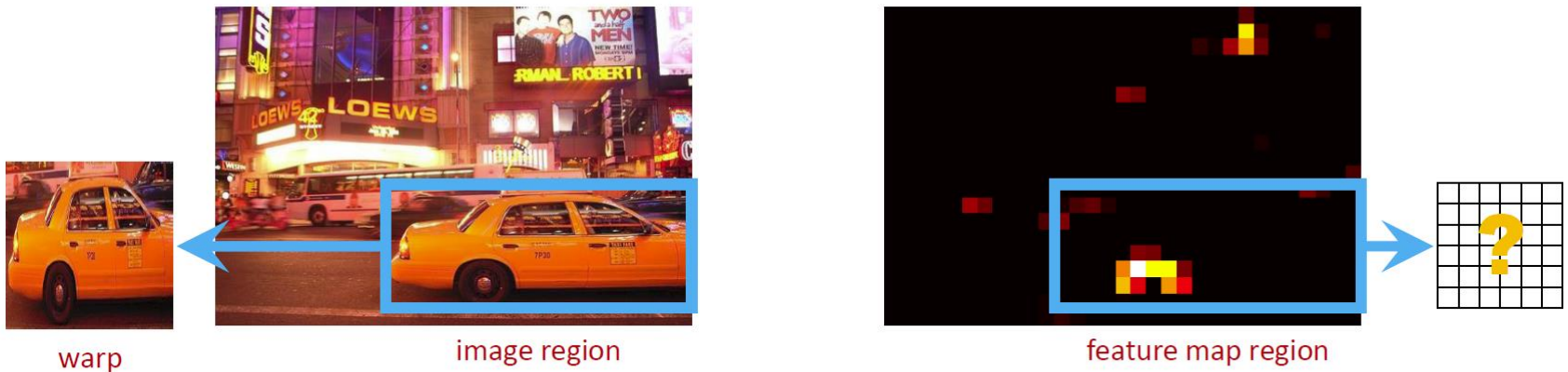
image region



feature map region

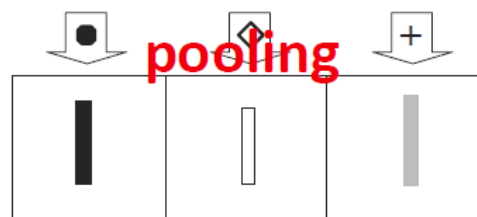
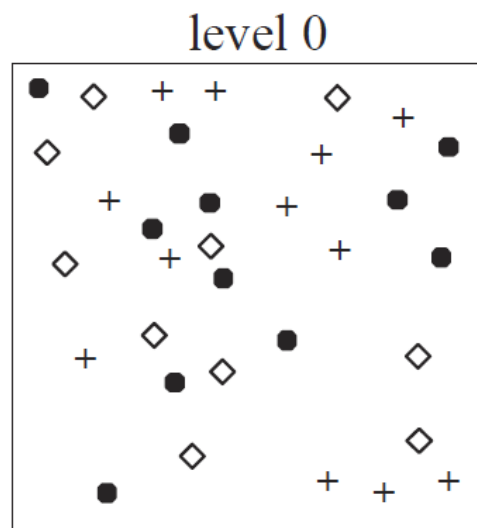
- Compute convolutional feature maps on the entire image **only once**.
- Project an image region to a **feature map region** (using correspondence of the receptive field center)
- Extract a region-based feature from the feature map region...

Regions on Feature Maps



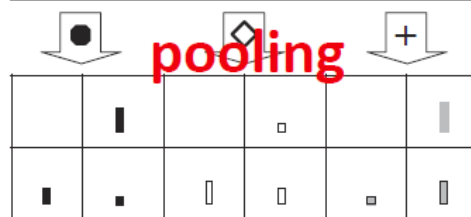
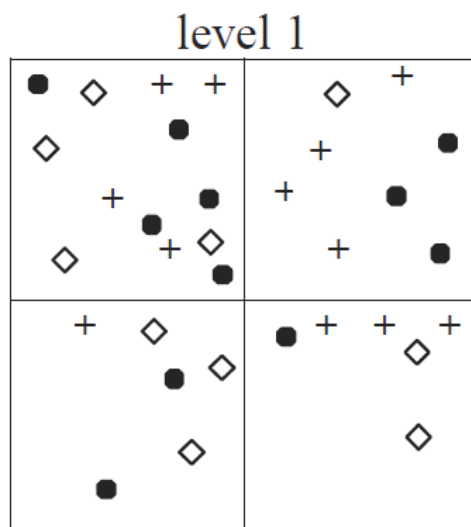
- **Fixed-length** features are required by fully-connected layers or SVM
- But how to produce a fixed-length feature from a feature map region?
- Solutions in traditional compute vision: Bag-of-words, SPM...

Bag-of-words & Spatial Pyramid Matching



Bag-of-words

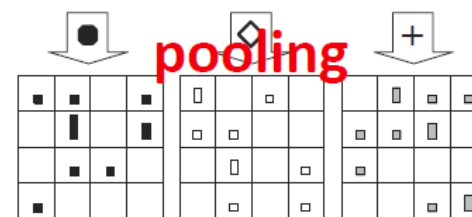
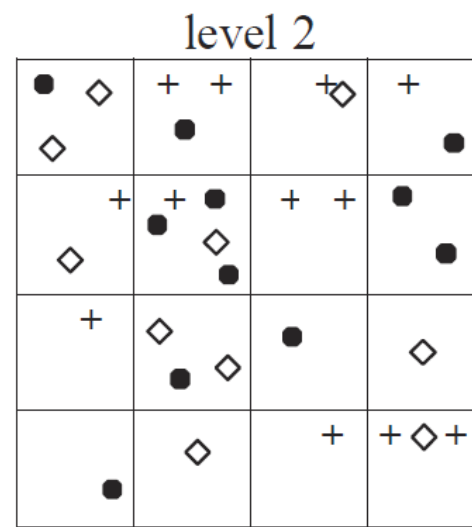
[J. Sivic & A. Zisserman, ICCV 2003]



Spatial Pyramid Matching (SPM)

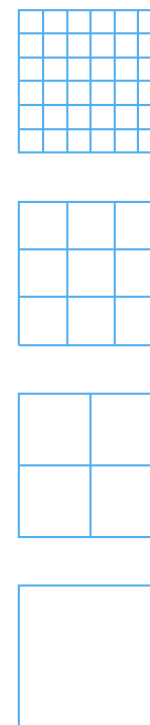
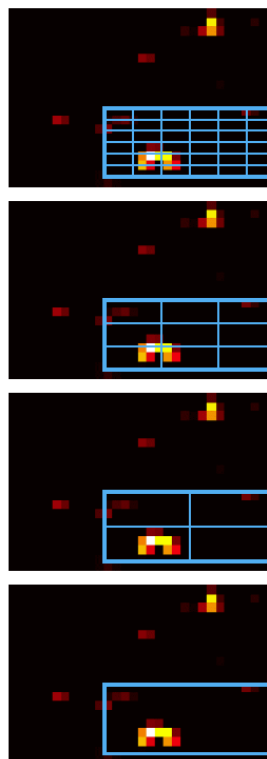
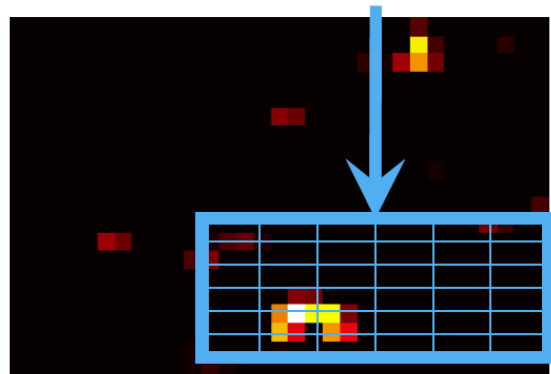
[K. Grauman & T. Darrell, ICCV 2005]

[S. Lazebnik et al, CVPR 2006]



Spatial Pyramid Pooling (SPP) Layer

- fix the number of bins (instead of filter sizes)
- **adaptively-sized** bins

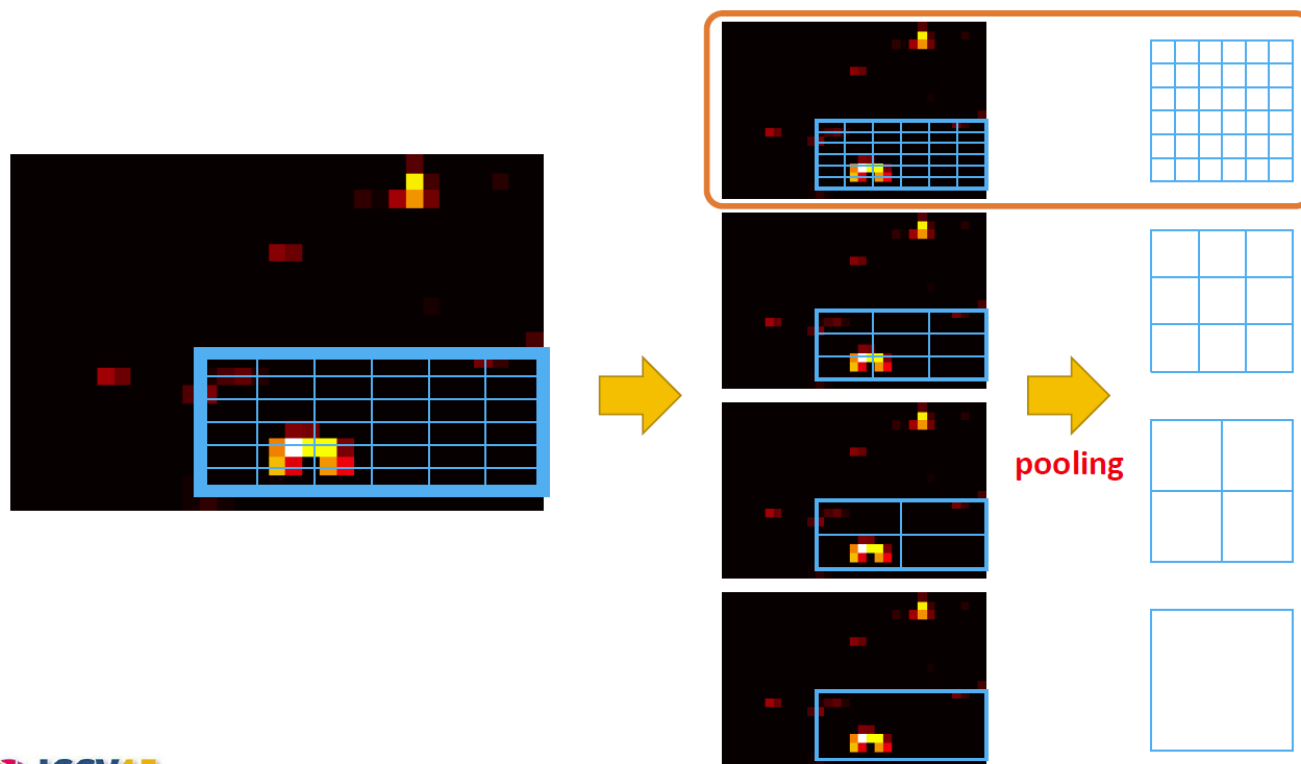


concatenate,
fc layers...

a finer level maintains
explicit spatial information

a coarser level removes
explicit spatial information
(bag-of-features)

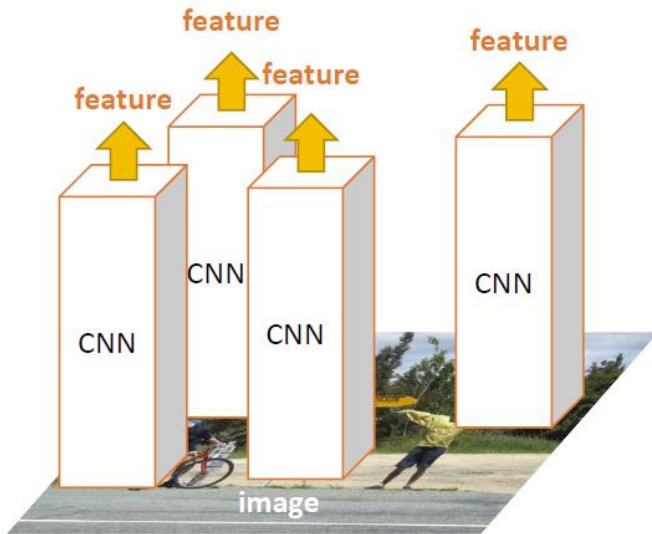
Spatial Pyramid Pooling (SPP) Layer



- Pre-trained nets often have a **single-resolution** pooling layer (7x7 for VGG nets)
- To adapt to a pre-trained net, a **“single-level”** pyramid is useable
- Region-of-Interest (RoI) pooling [R. Girshick, ICCV 2015]

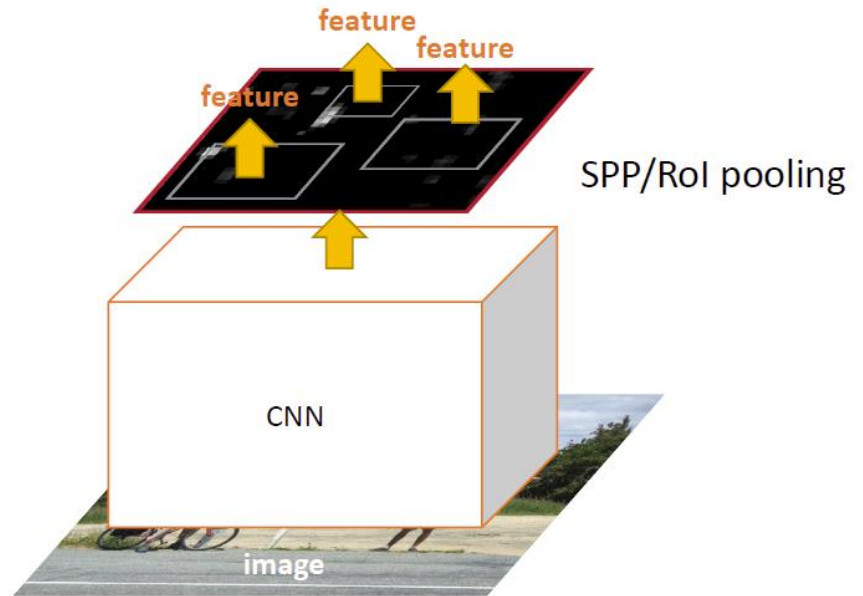
concatenate,
fc layers...

RCNN v.s. Fast RCNN



R-CNN

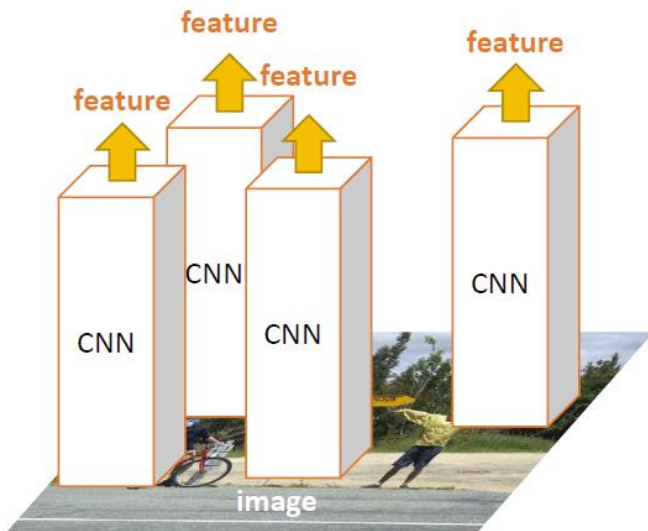
- Extract image regions
- 1 CNN per region (2000 CNNs)
- Classify region-based features



SPP-net & Fast R-CNN (the same forward pipeline)

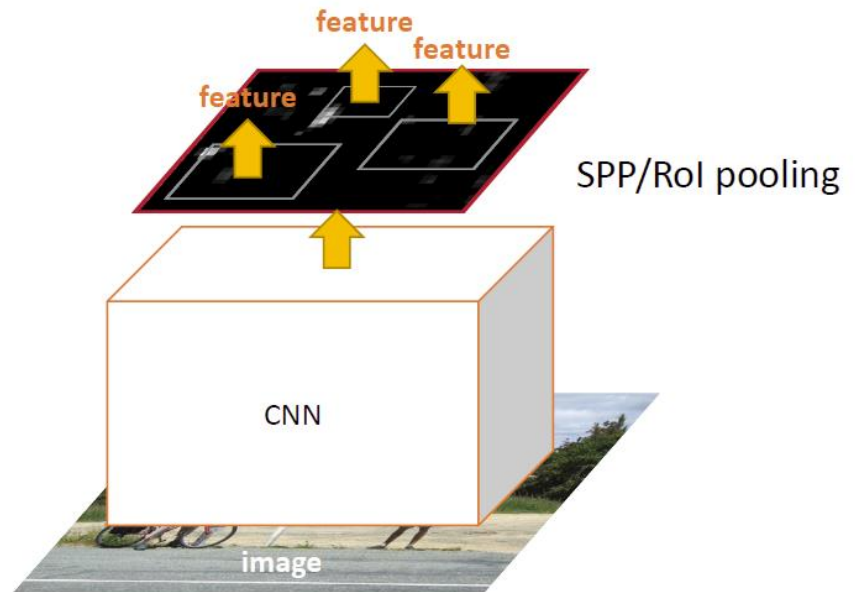
- 1 CNN on the entire image
- Extract features from feature map regions
- Classify region-based features

RCNN v.s. Fast RCNN



R-CNN

- Complexity: $\sim 224 \times 224 \times 2000$



SPP-net & Fast R-CNN (the same forward pipeline)

- Complexity: $\sim 600 \times 1000 \times 1$
- $\sim 160\times$ faster than R-CNN

Region Proposal from Feature Maps

- Object detection networks are fast (0.2s)...
- but what about region proposal?
- Selective Search [Uijlingset al. ICCV 2011]: 2s per image
- EdgeBoxes[Zitnick & Dollar. ECCV 2014]: 0.2s per image
- Can we do region proposal on the same set of feature maps?

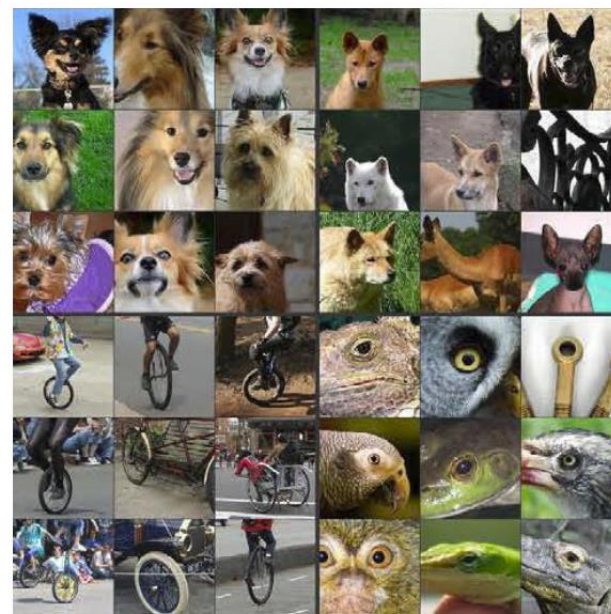
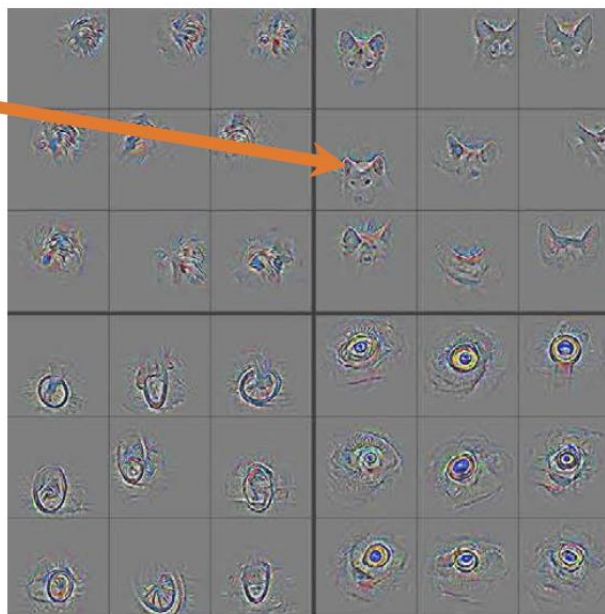
Shaoqing Ren, Kaiming He, Ross Girshick, & Jian Sun. "Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks". NIPS 2015.

Region Proposal from Feature Maps

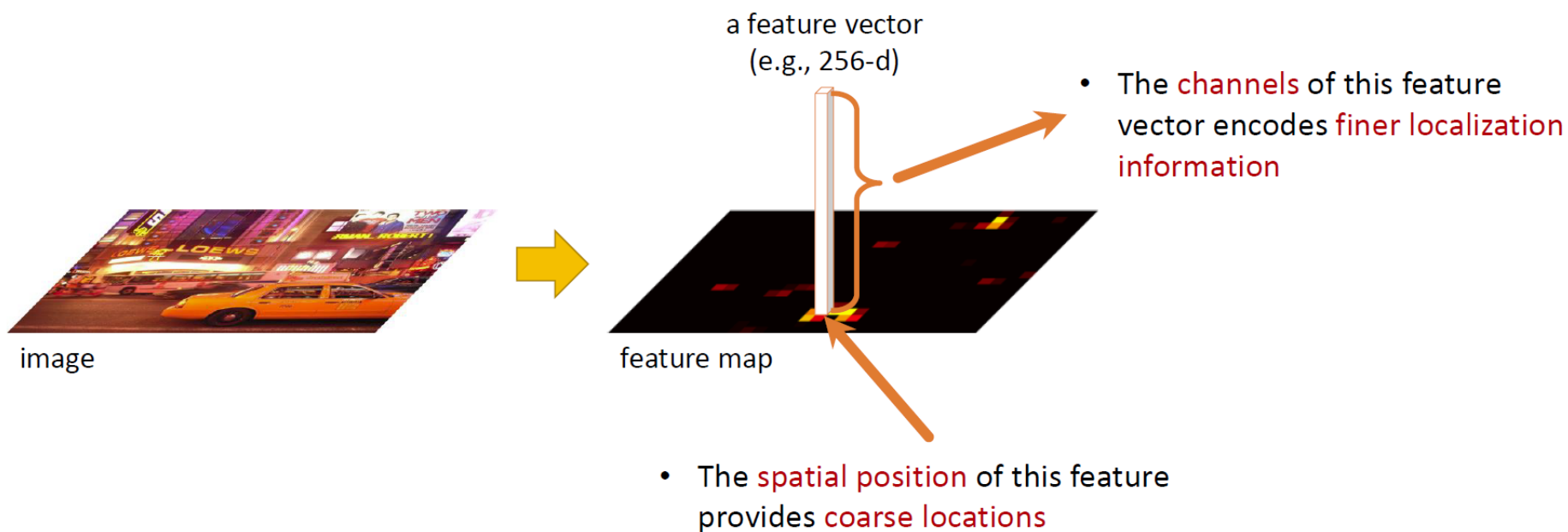
- By decoding **one response** at a single pixel, we can still roughly see the object outline*
- **Finer localization information** has been encoded in the channels of a convolutional feature response
- Extract this information for better localization...

* Zeiler & Fergus's method traces unpooling information so the visualization involves more than a single response. But other visualization methods reveal similar patterns.

Revisiting visualizations from Zeiler & Fergus

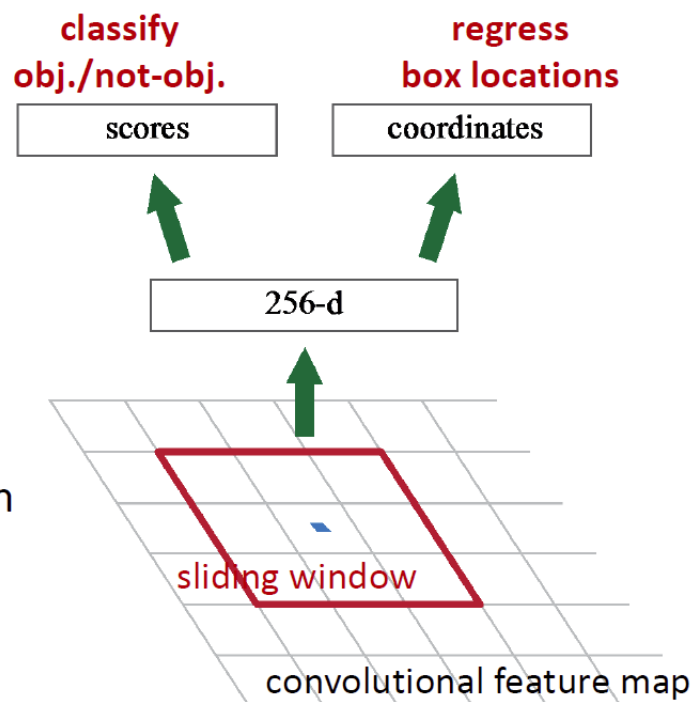


Region Proposal from Feature Maps



Region Proposal Network

- Slide a small window on the feature map
- Build a small network for:
 - classifying object or not-object, and
 - regressing bbox locations
- Position of the sliding window provides localization information **with reference to the image**
- Box regression provides finer localization information **with reference to this sliding window**



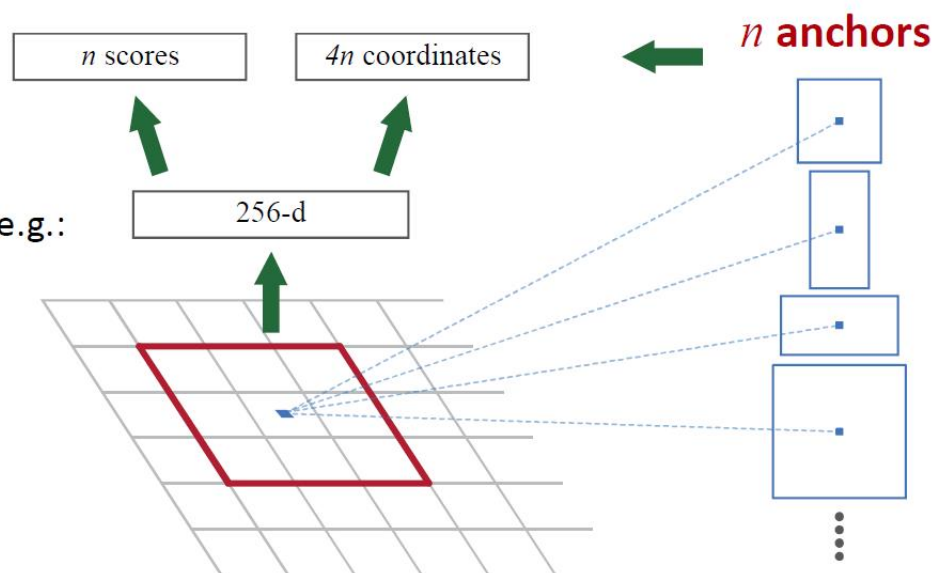
Anchors as references

- **Anchors:** pre-defined reference boxes

- Box regression is with reference to anchors: regressing an anchor box to a ground-truth box

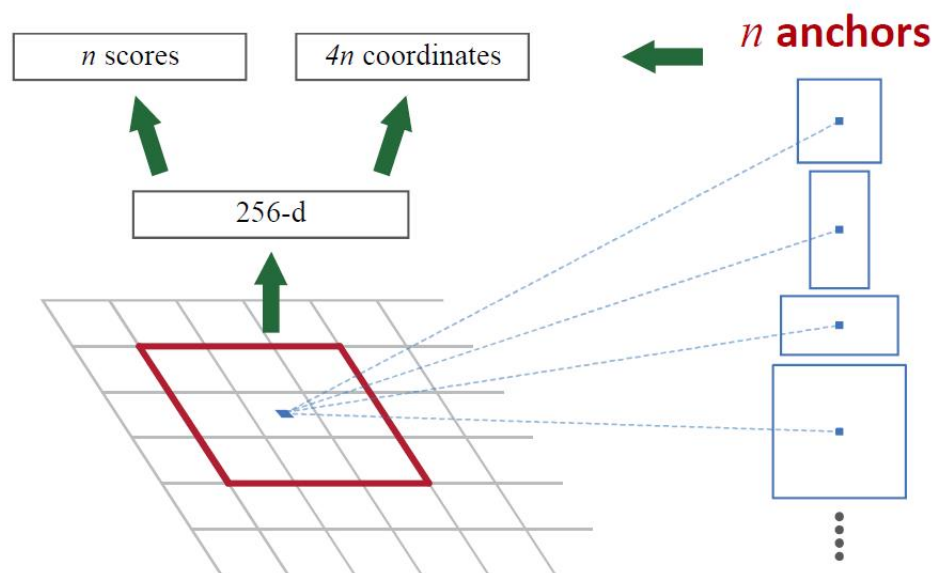
- Object probability is with reference to anchors, e.g.:

- anchors as positive samples: if IoU > 0.7 or IoU is max
- anchors as negative samples: if IoU < 0.3



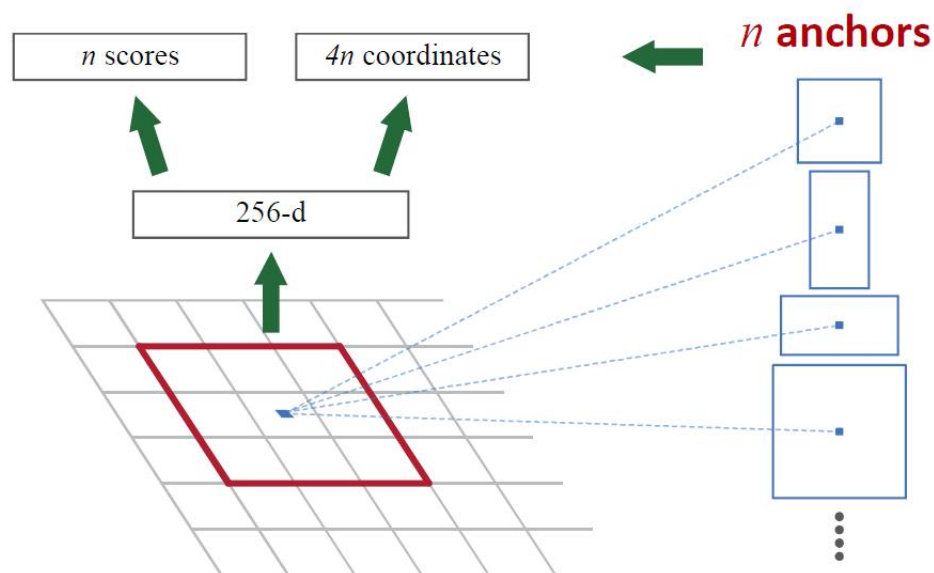
Anchors as references

- **Anchors**: pre-defined reference boxes
- **Translation-invariant** anchors:
 - the same set of anchors are used at each sliding position
 - the same prediction functions (with reference to the sliding window) are used
 - a translated object will have a translated prediction



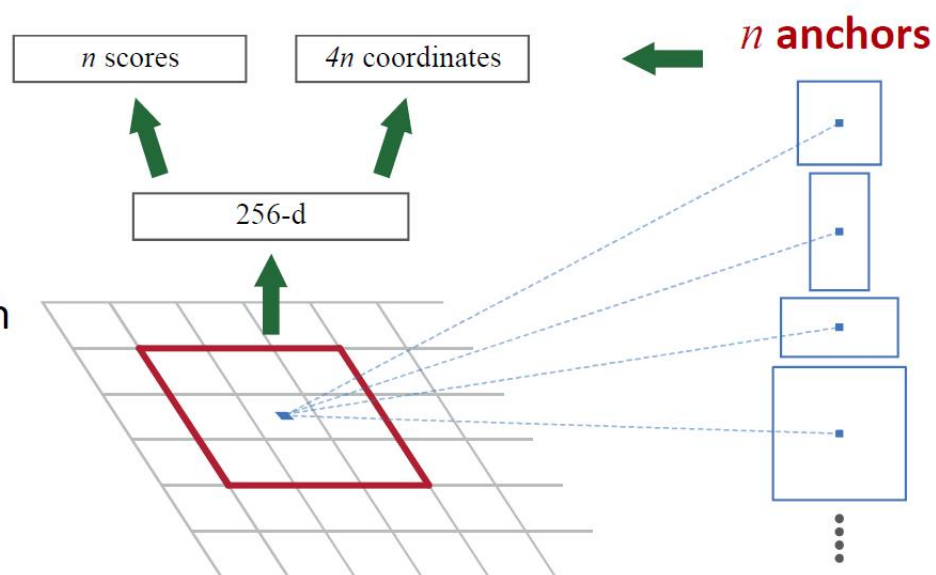
Anchors as references

- **Anchors**: pre-defined reference boxes
- **Multi-scale/size** anchors:
 - multiple anchors are used at each position: e.g., 3 scales (128^2 , 256^2 , 512^2) and 3 aspect ratios (2:1, 1:1, 1:2) yield 9 anchors
 - each anchor has its own prediction function
 - **single-scale** features, multi-scale predictions

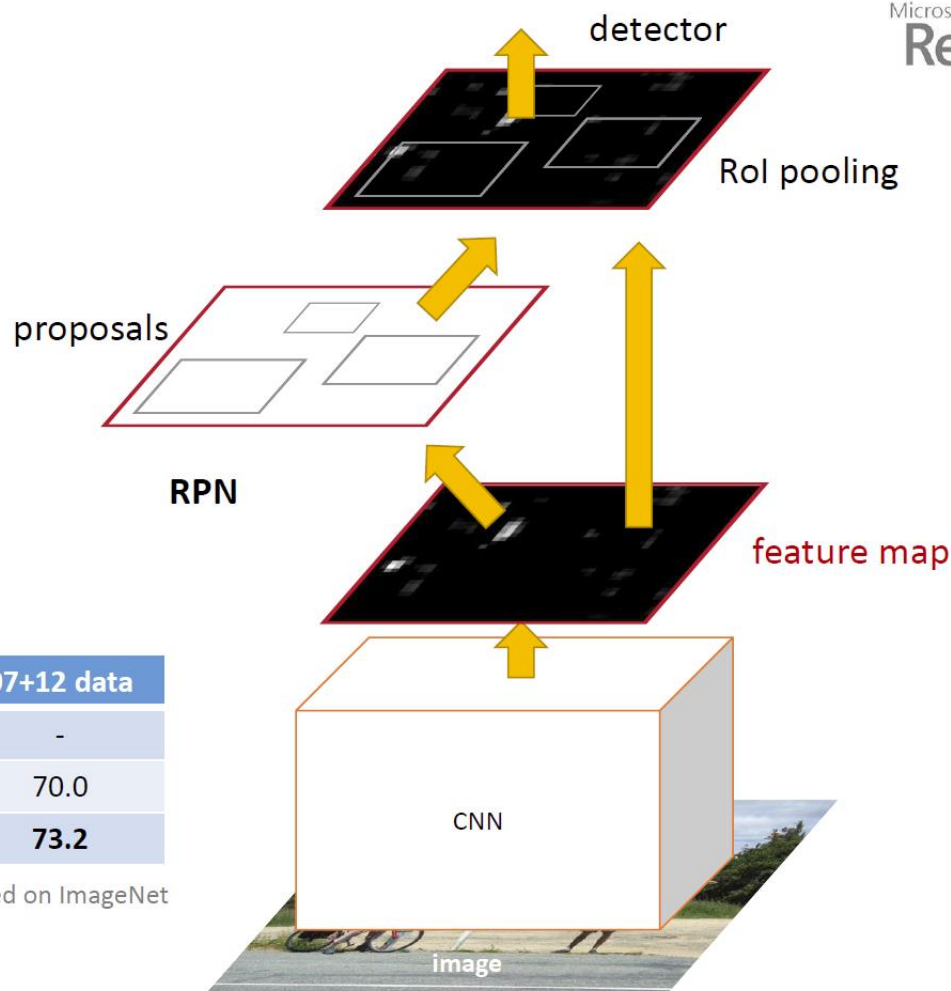


Region Proposal Network

- RPN is **fully convolutional** [Long et al. 2015]
- RPN is trained end-to-end
- RPN **shares** convolutional feature maps with the detection network
(covered in Ross's section)



Faster R-CNN



system	time	07 data	07+12 data
R-CNN	~50s	66.0	-
Fast R-CNN	~2s	66.9	70.0
Faster R-CNN	198ms	69.9	73.2

detection mAP on PASCAL VOC 2007, with VGG-16 pre-trained on ImageNet

Bounding Box Regression

- Input: A set of N training pairs $\{(P^i, G^i)\}_{i=1,\dots,N}$, where $P^i = (P_x^i, P_y^i, P_w^i, P_h^i)$ and $G^i = (G_x^i, G_y^i, G_w^i, G_h^i)$. (Drop superscript i for simplicity)
- Output: Four functions $d_x(P), d_y(P), d_w(P), d_h(P)$

$$\hat{G}_x = P_w d_x(P) + P_x \quad (1)$$

$$\hat{G}_y = P_h d_y(P) + P_y \quad (2)$$

$$\hat{G}_w = P_w \exp(d_w(P)) \quad (3)$$

$$\hat{G}_h = P_h \exp(d_h(P)). \quad (4)$$

- Learn model parameters by optimizing the regularized least squares objective

$$\mathbf{w}_\star = \operatorname{argmin}_{\hat{\mathbf{w}}_\star} \sum_i^N (t_\star^i - \hat{\mathbf{w}}_\star^\top \phi_5(P^i))^2 + \lambda \|\hat{\mathbf{w}}_\star\|^2. \quad (5)$$

Bounding Box Regression

- Learn model parameters by optimizing the regularized least squares objective

$$\mathbf{w}_\star = \underset{\hat{\mathbf{w}}_\star}{\operatorname{argmin}} \sum_i^N (t_\star^i - \hat{\mathbf{w}}_\star^\top \phi_5(P^i))^2 + \lambda \|\hat{\mathbf{w}}_\star\|^2. \quad (5)$$

- The regression targets $t_\star$ for the training pair (P, G) are defined as

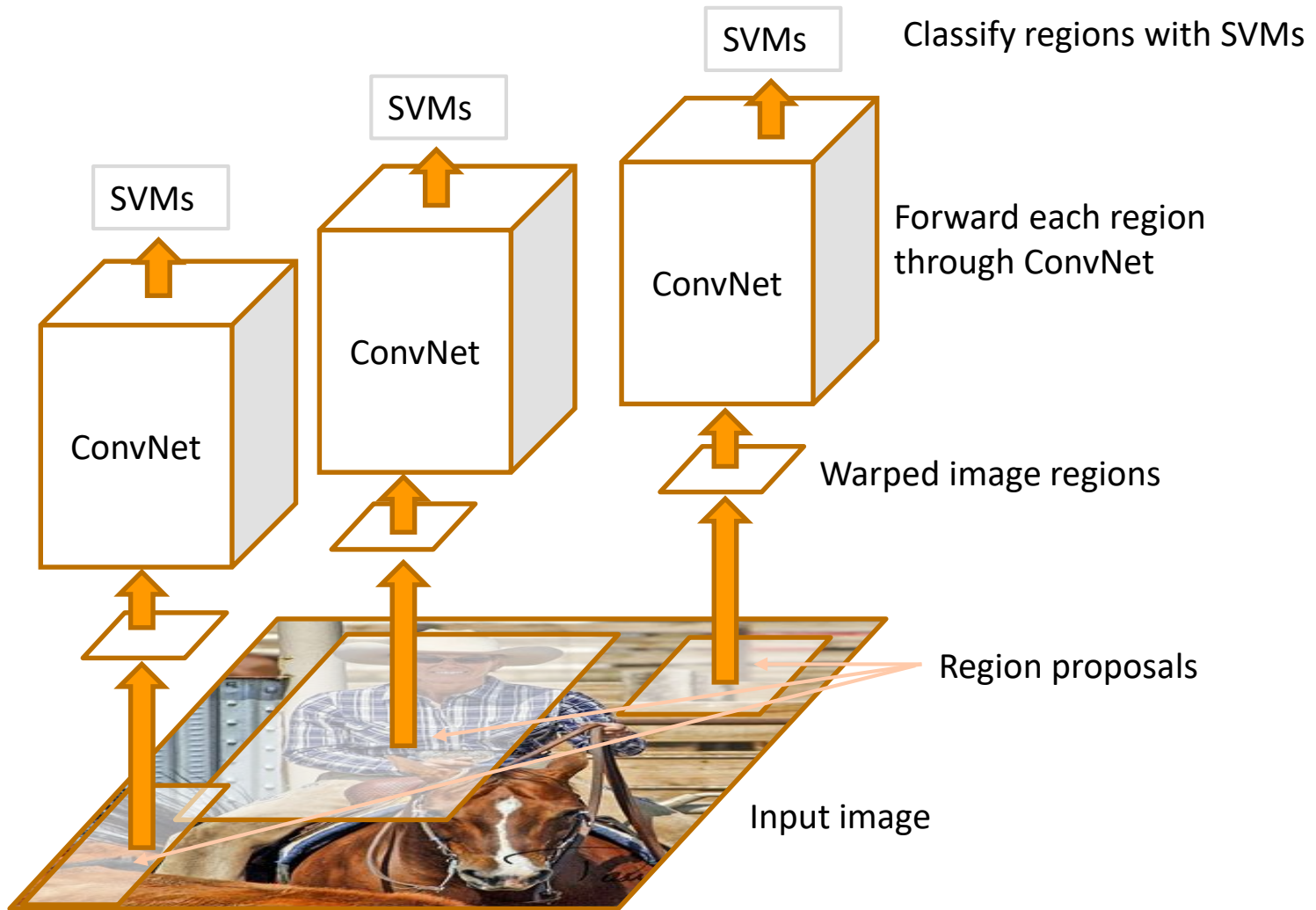
$$t_x = (G_x - P_x)/P_w \quad (6)$$

$$t_y = (G_y - P_y)/P_h \quad (7)$$

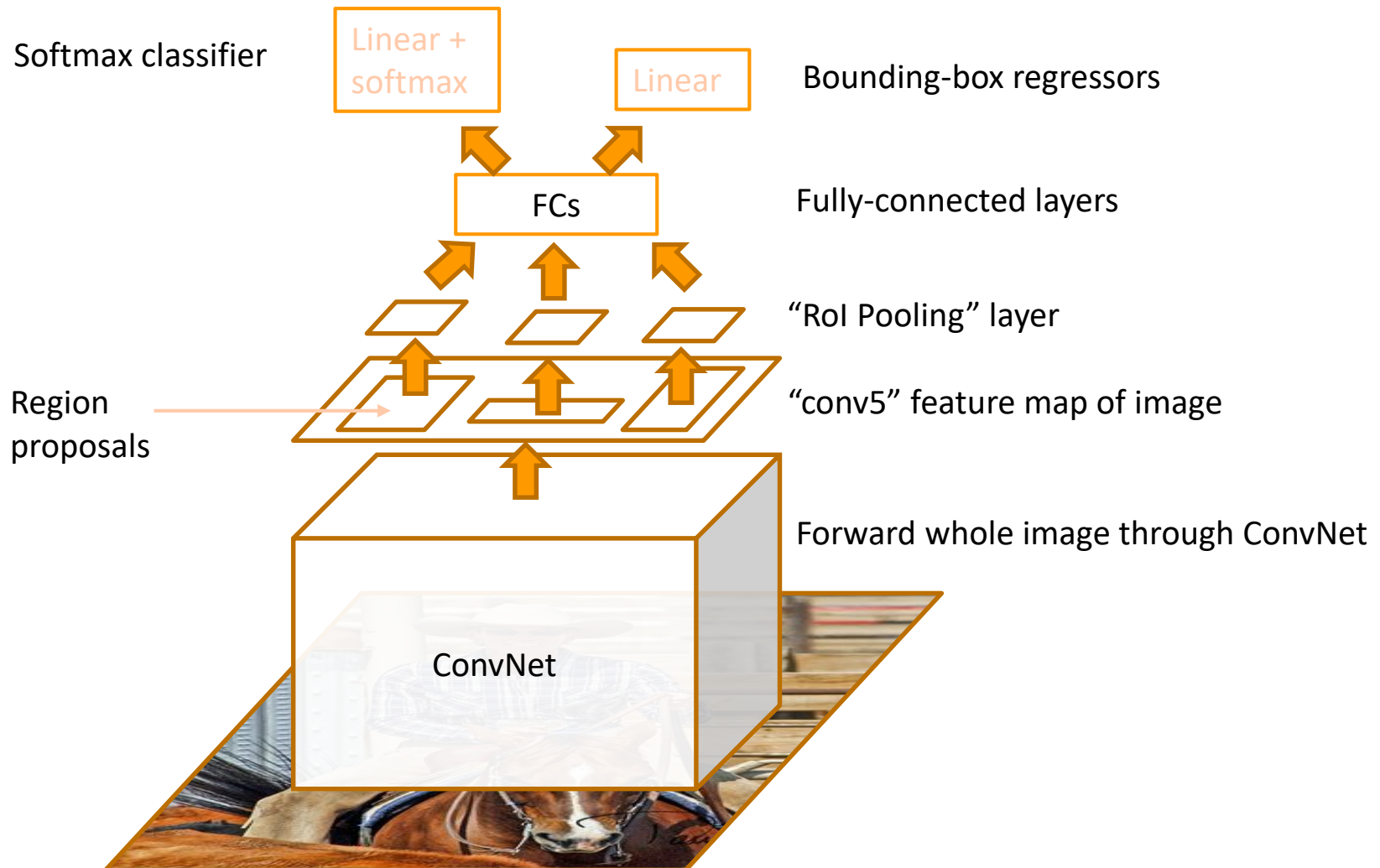
$$t_w = \log(G_w/P_w) \quad (8)$$

$$t_h = \log(G_h/P_h). \quad (9)$$

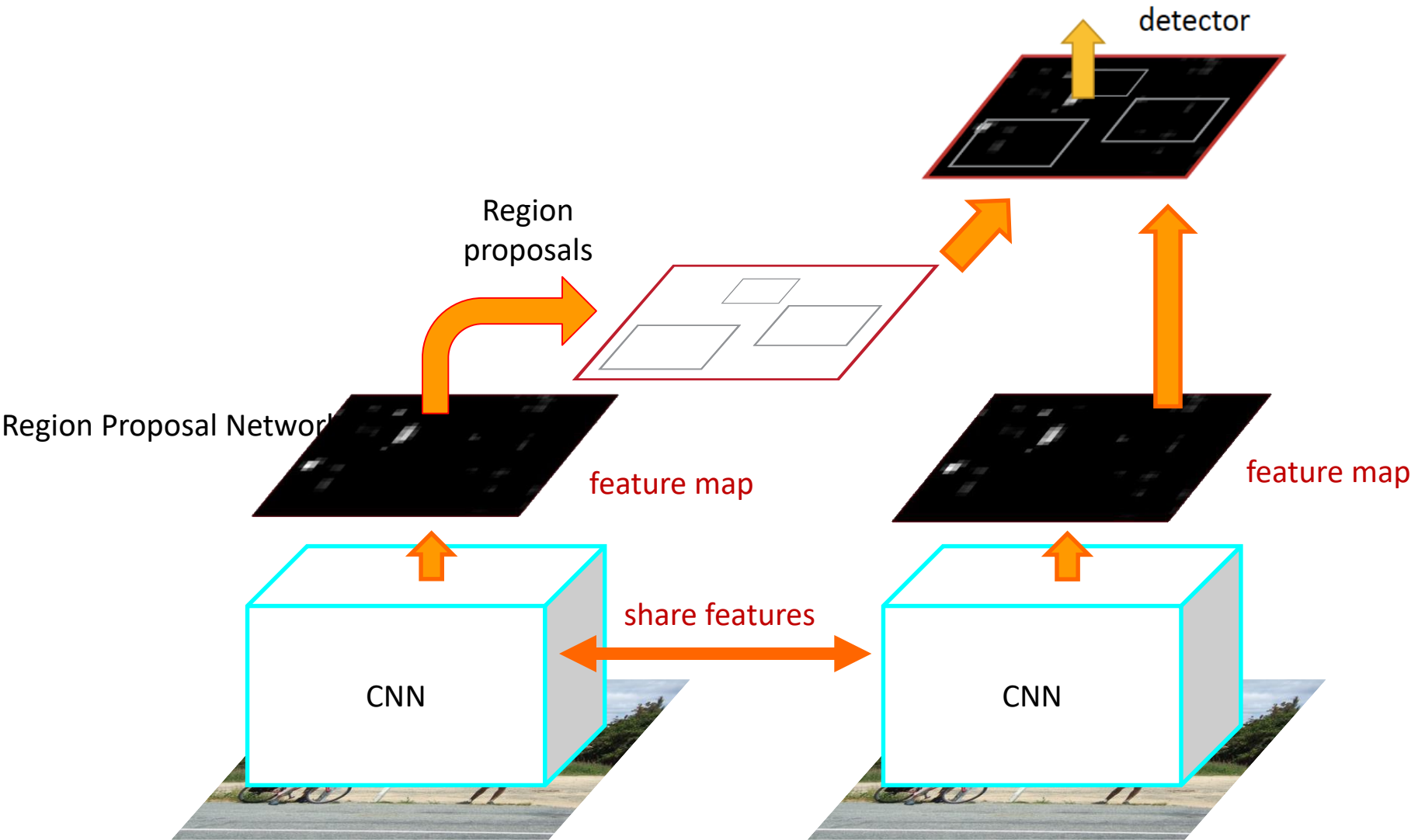
Review: R-CNN



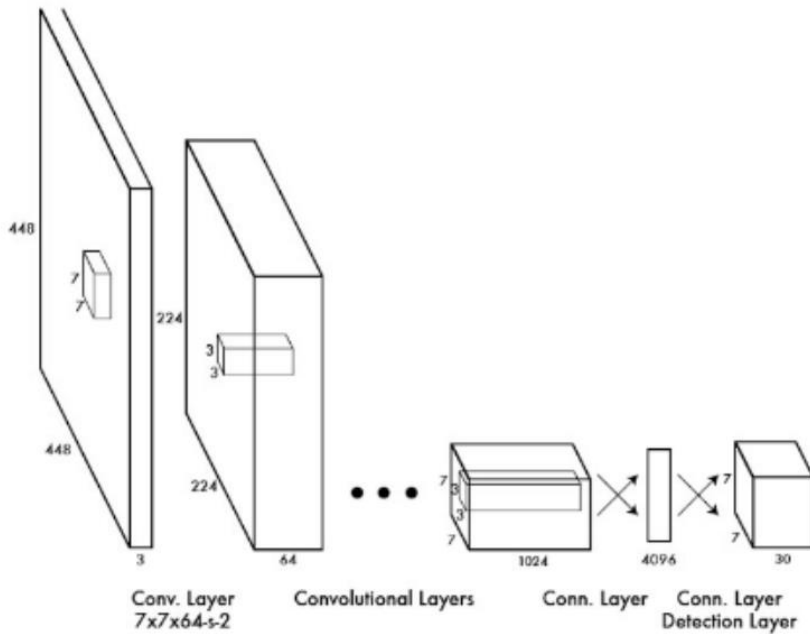
Review: Fast R-CNN



Review: Faster R-CNN

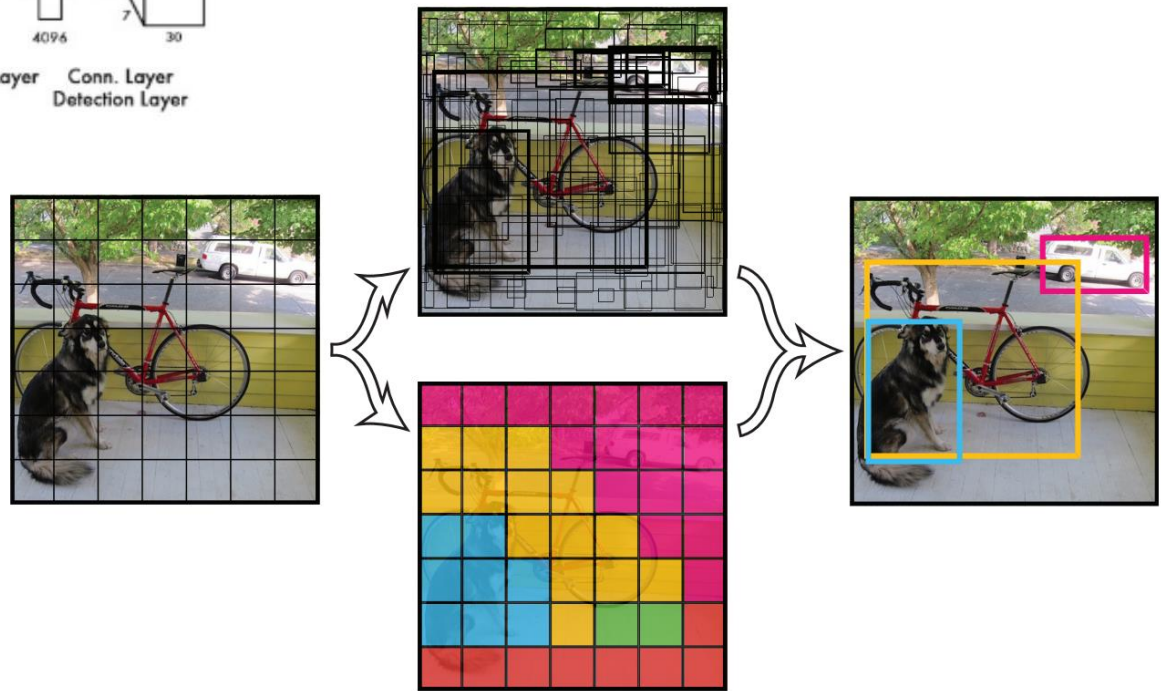


YOLO



Regression instead of classification:

If the center of an object falls into a grid cell, that grid cell is responsible for detecting that object.

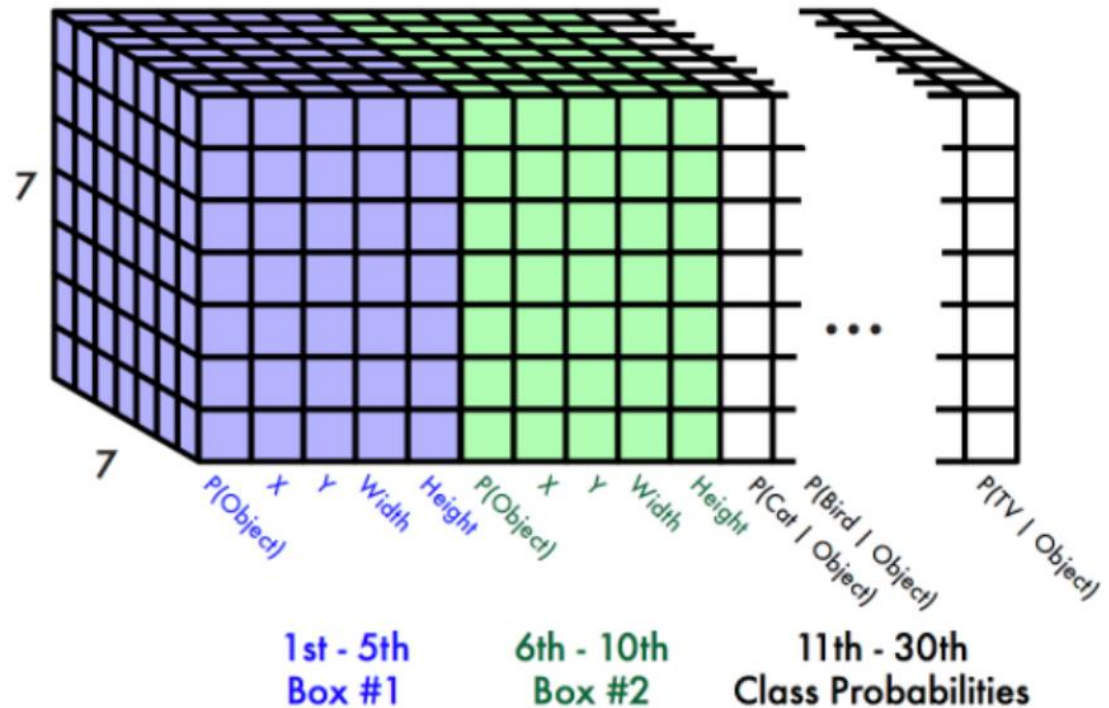


Redmon J, et al. You only look once: Unified, real-time object detection. CVPR2016

YOLO

Each cell predicts:

- For each bounding box:
 - 4 coordinates (x, y, w, h)
 - 1 confidence value
- Some number of class probabilities



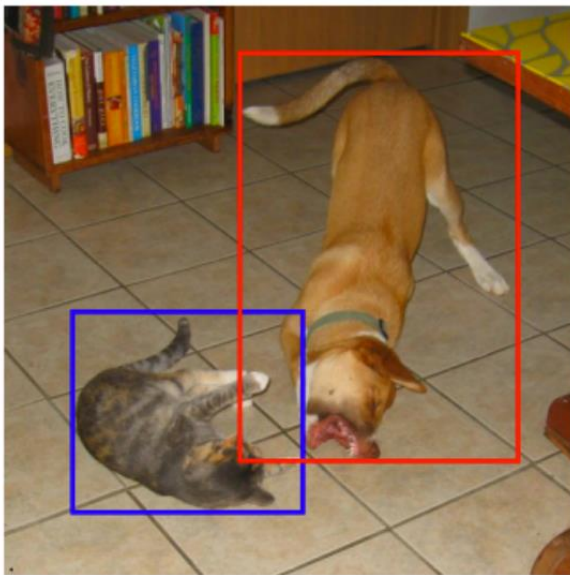
For Pascal VOC:

- 7x7 grid
- 2 bounding boxes / cell
- 20 classes

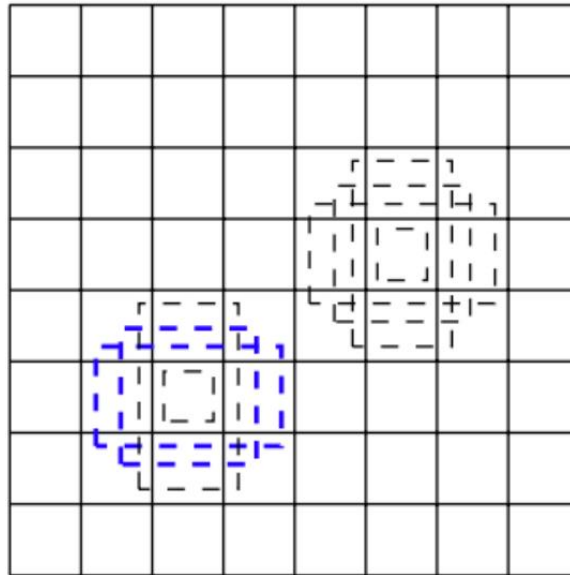
$7 \times 7 \times (2 \times 5 + 20) = 7 \times 7 \times 30$ tensor = **1470 outputs**

SSD

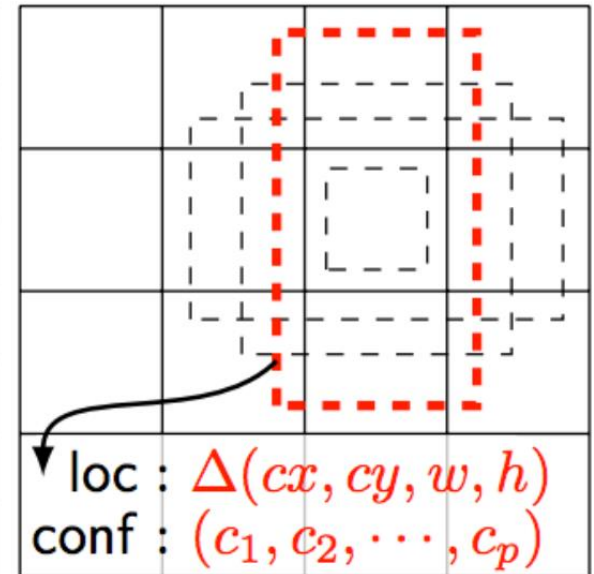
- SSD: YOLO + default box shape + multi-scale



(a) Image with GT boxes



(b) 8×8 feature map

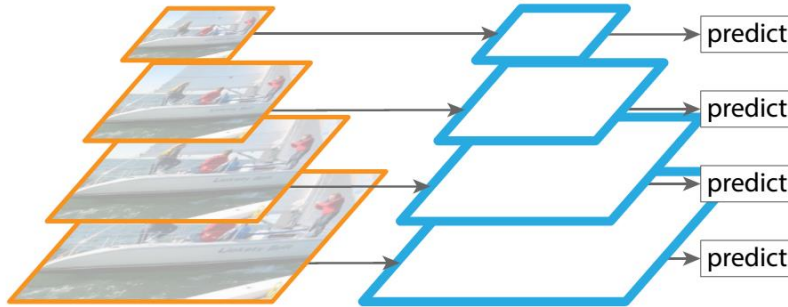


loc : $\Delta(cx, cy, w, h)$
conf : $(c_1, c_2, \dots, c_p)$

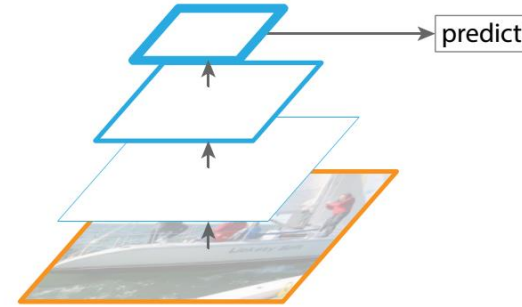
(c) 4×4 feature map

FPN (Feature Pyramid Network)

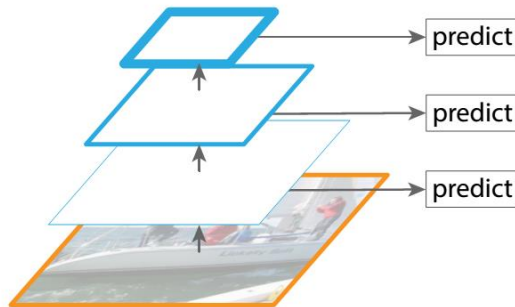
- Explore the power of multiple scales



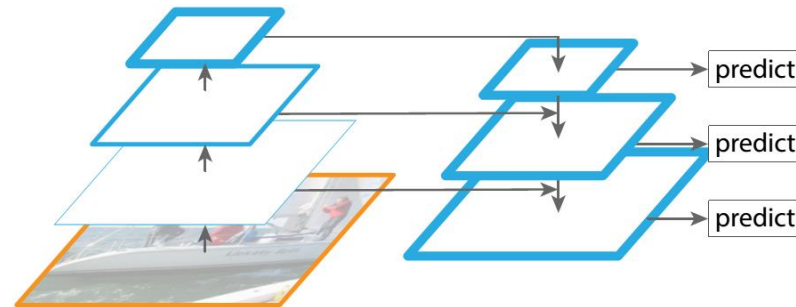
(a) Featurized image pyramid



(b) Single feature map



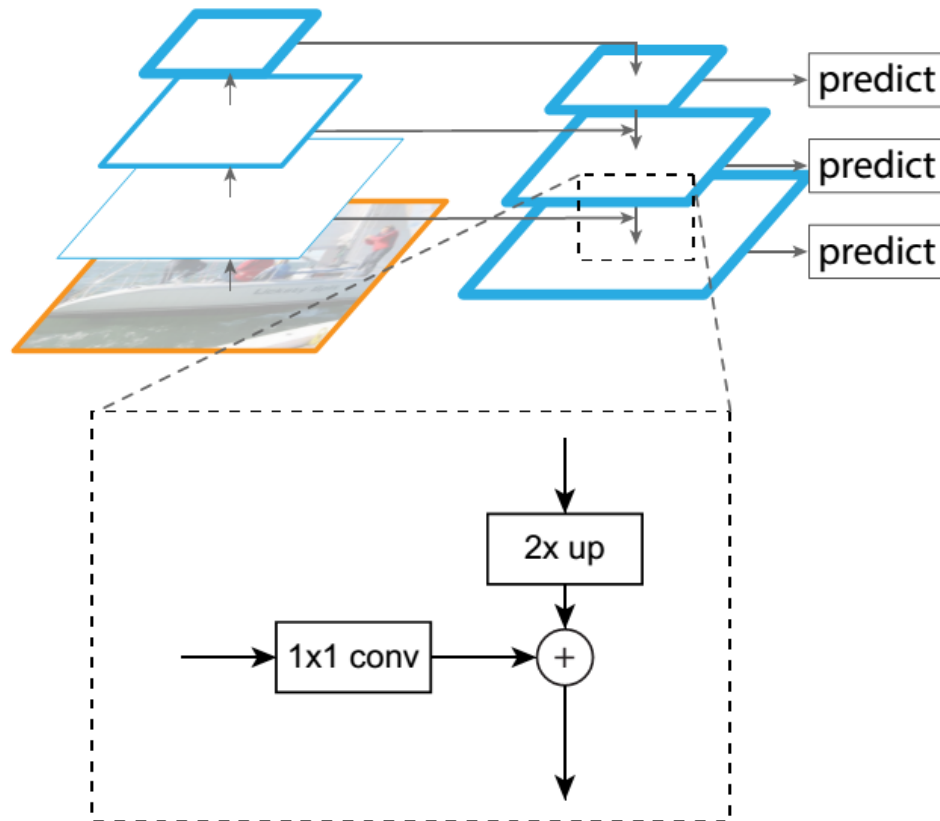
(c) Pyramidal feature hierarchy



(d) Feature Pyramid Network

FPN (Feature Pyramid Network)

- The top-down pathway



Question?